



Typical Service Implementation Solutions on the Cloud



Foreword

- An application instance is the minimum unit for deploying services on the cloud. This lesson describes how to deploy application instances on Huawei Cloud, which Huawei Cloud resources should be used, and what are the optimization and evolution strategies for service systems deployed on Huawei Cloud.



Objectives

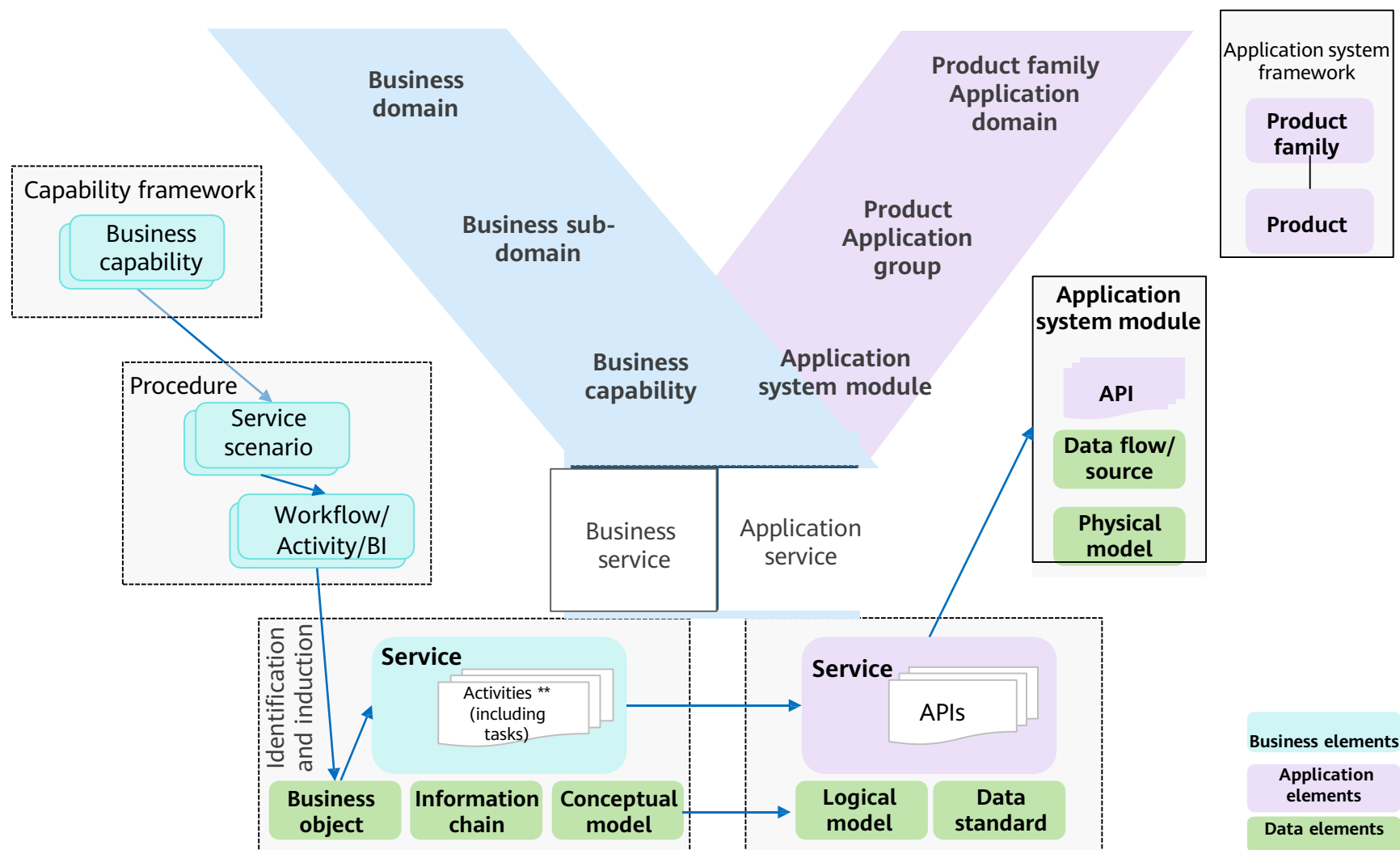
- Upon completion of this course, you will understand:
 - Common application architectures
 - Infrastructure services of Huawei Cloud
 - Evolution strategies of the Huawei Cloud HA architecture



Contents

- 1. Typical Service System Construction on the Cloud**
2. Discussion on Cloud-based HA Architecture Design

Application Definition



Application: a logical set of one or more independent and complete functions that are closely related to each other and have specific business features. Application services provide capabilities for external systems through reusable APIs.

Business: a way in which the values of business capabilities are presented. Business services are goods or services provided by enterprises or organizations through explicit APIs, and are explicitly managed by organizations.

Applications and Application Instances

An application is the smallest unit of software that can independently complete a service process for a customer. An application must have the following two elements:

- Only one vendor: The vendor is responsible for any application code changes.
- Only one owner: The owner on the customer side is responsible for the lifecycle of the application, including the upgrade, update, replacement, and elimination of the application. They have the right to decide when to cut over applications and are responsible for application acceptance.

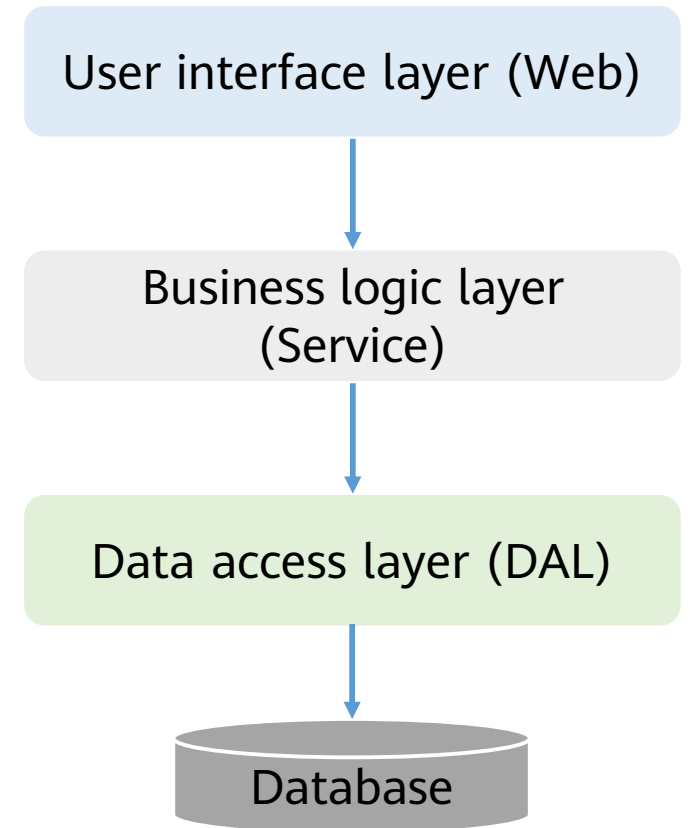
An application consists of a set of software systems and has a unique vendor and owner without any specific operating environment attributes.

An application instance is an independent operating environment for an application. It has specific operating environment attributes, including the account and password, access address, and operating environmental variables.

- Ratio between applications and application instances: 1: N
- Minimum unit of application cloud migration: application instance

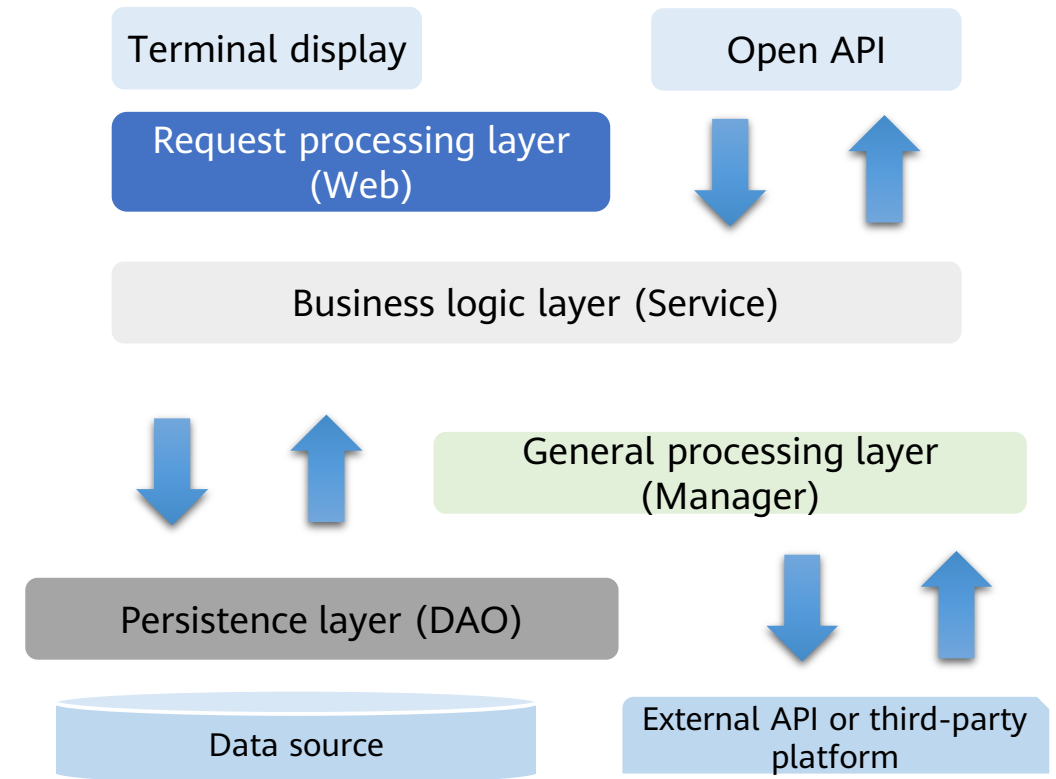
Application Layering: Typical Three-Layer Application Architecture

- A typical three-layer application architecture consists of, from top to bottom, the user interface layer, business logic layer, and data access layer.
 - User interface layer (UIL): This layer is also called the web layer.
 - Business logic layer (BLL): This layer is also called the domain layer or service layer, which processes system service logic.
 - Data access layer (DAL): This layer directly communicates with the database by allowing for operations such as adding, deleting, modifying, updating, and querying data. This layer is also called the data access object (DAO) pattern.



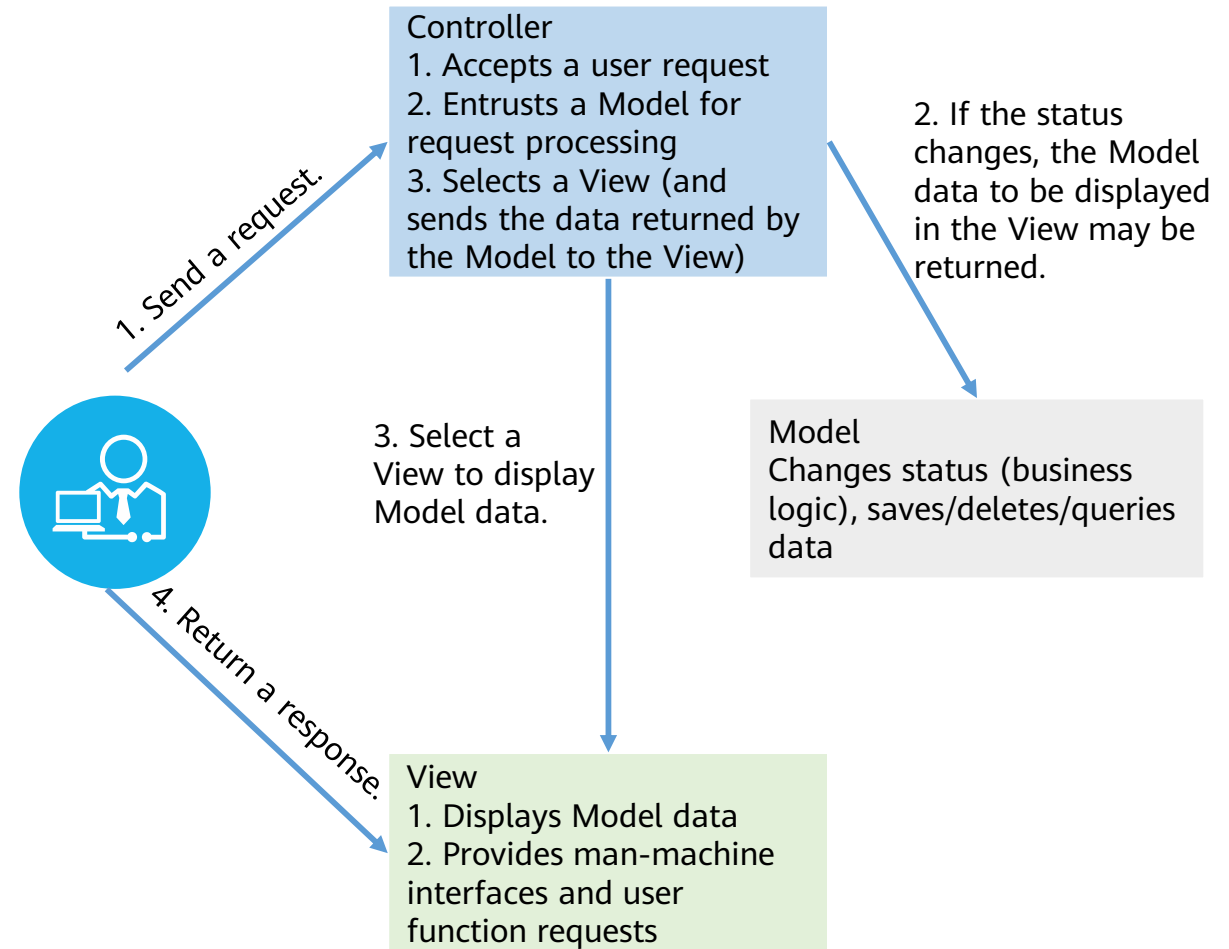
Application Layering: Four-Layer Architecture

- The implementation of the three-layer architecture is simple, but it is prone to bloat as business logic tends to concentrate at the business logic layer. A four-layer architecture looks to refine request processing by adding a general processing layer (manager layer) to the original three-layer architecture.
 - **Web layer:** forwards access control, verifies basic parameters, and processes services that are not reused
 - **Service layer:** a specific business logic layer
 - **Manager layer:** general service processing layer, which has the following features:
 - Pre-processes the returned results and information about abnormal conversion for the layer encapsulated by the third-party platform
 - Provides common capabilities of the service layer, such as the cache solution and middleware processing
 - Interacts with the DAO layer and combines and reuses multiple DAOs
 - **DAO layer:** data access layer, which exchanges data with the underlying MySQL, Oracle, and HBase

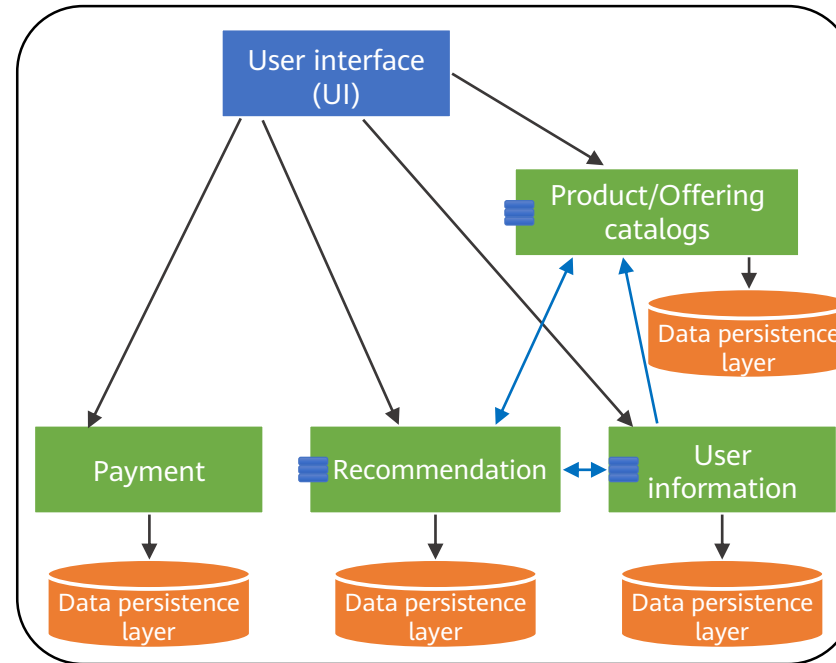


Application Layering: MVC Architecture

- The Model-View-Controller (MVC) is an architectural pattern that separates an application into three main logical components: the Model, the View, and the Controller. It aggregates business logic into a component so that developers do not need to rewrite business logic while improving and personalizing interfaces and user interaction.
 - View: provides a GUI for users to interact with each other
 - Model: carries data and processes user requests. This component is responsible for processing application data logic, for example, accessing databases.
 - Controller: forwards a user request to a corresponding Model component for processing, processes the calculation result given by the Model, and provides a response to the user



Application Layering: Microservice Architecture



- The microservice architecture breaks down applications into **a group of small services** called microservices, which work with each other to provide business capabilities.
- Each microservice is **designed, developed, and tested by an independent team**, and can be deployed separately in the production environment.
- Each microservice has its specific functions. Specific microservices are selected to implement specific functions of an application.

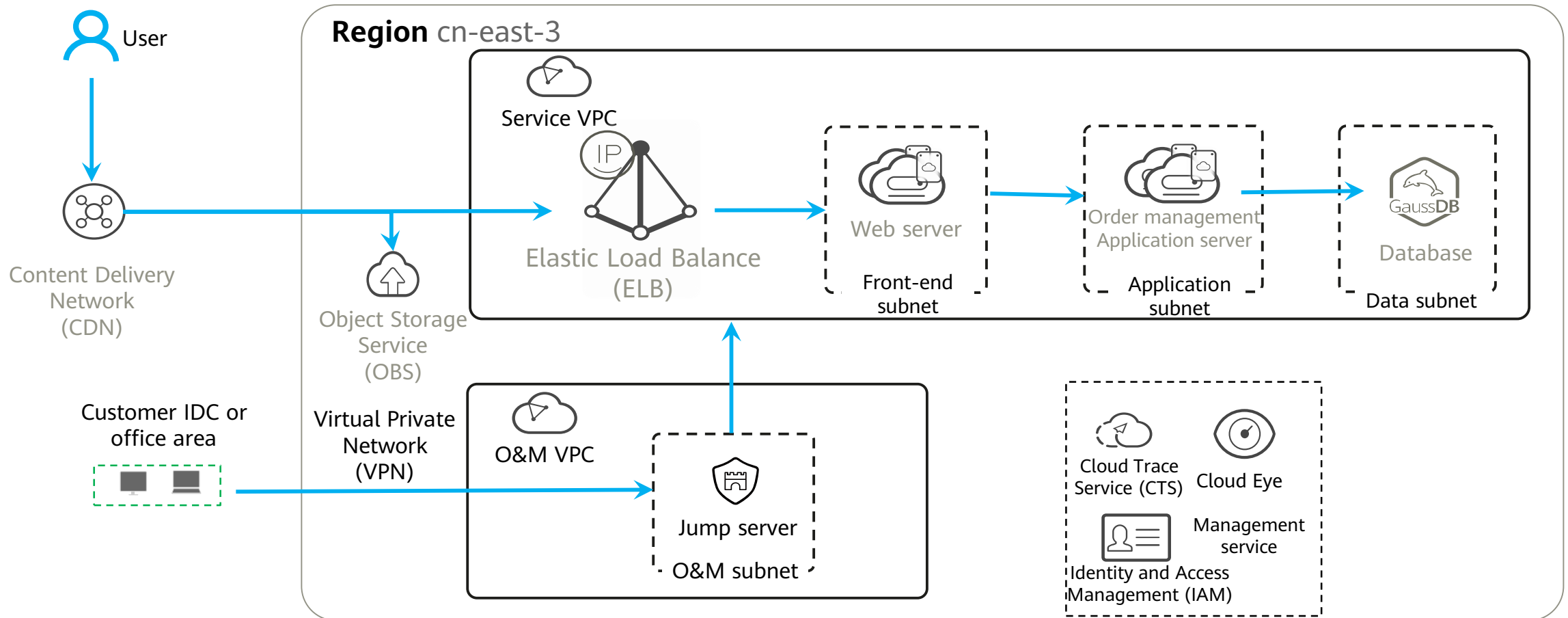
Q&A



If you wanted to deploy an online shopping platform on Huawei Cloud, how would you design the three-layer architecture?

Architecture for Service Implementation on the Cloud (Example)

- The following uses a typical three-layer architecture as an example to describe the architecture of an online shopping platform deployed on the cloud.



Service List

Service	Description
ECS	An Elastic Cloud Server (ECS) is a basic computing component that consists of vCPUs, memory, OS, and Elastic Volume Service (EVS) disks. After an ECS is created, you can use it like using your local computer or physical server.
VPC	The Virtual Private Cloud (VPC) service enables you to provision logically isolated and private virtual networks for cloud resources, such cloud servers, containers, and databases. You can customize subnets, security groups, network ACLs, and assign EIPs and bandwidths. With Direct Connect or Virtual Private Network (VPN), you can connect your VPCs to an on-premises data center.
Security Group	A security group is a collection of access control rules for cloud resources, such as cloud servers, containers, and databases, that have the same security requirements and that are mutually trusted. After a security group is created, you can create various access rules for the security group and these rules will apply to all cloud resources added to this security group.
Subnet	A subnet is a unique CIDR block with a range of IP addresses in a VPC. All resources in a VPC must be deployed on subnets.
Network ACL	A network ACL is an optional layer of security for your subnets. After you associate one or more subnets with a network ACL, you can control traffic in and out of the subnets.
EIP	The Elastic IP (EIP) service provides independent public IP addresses and bandwidth for Internet access. After an EIP is configured for a resource, the resource can directly access the Internet. If only a private IP address is configured for the resource, the resource cannot directly access the Internet. You can easily bind an EIP to an ECS, BMS, virtual IP address, load balancer, or NAT gateway, enabling immediate Internet access.
RDS	Relational Database Service (RDS) is a cloud-based web service that is reliable, scalable, easy to manage, and immediately ready for use.
OBS	Object Storage Service (OBS) is a cloud storage service optimized for storing massive amounts of data. It provides unlimited, secure, and highly reliable storage capabilities at a low cost.
CTS	Cloud Trace Service (CTS) records operations on the cloud resources in your account. You can use the logs generated by CTS to perform security analysis, track resource changes, audit compliance, and locate faults.
CES	Cloud Eye is a multi-dimensional monitoring platform that monitors resources such as ECSs and bandwidth. You can use Cloud Eye to monitor resources, set alarm rules, and identify resource exceptions to quickly handle resource changes.
IAM	Identity and Access Management (IAM) is a basic service of Huawei Cloud that provides permissions management to help you securely control access to your cloud services and resources.

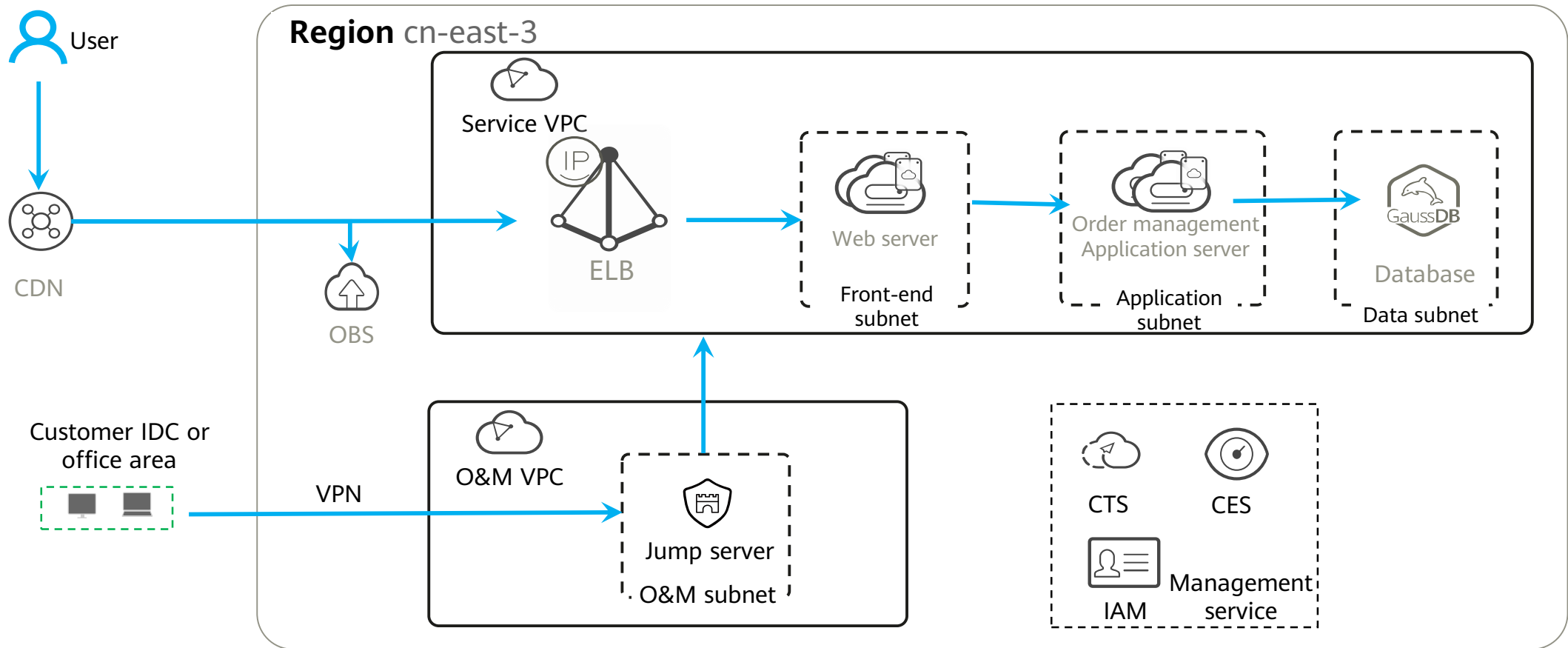


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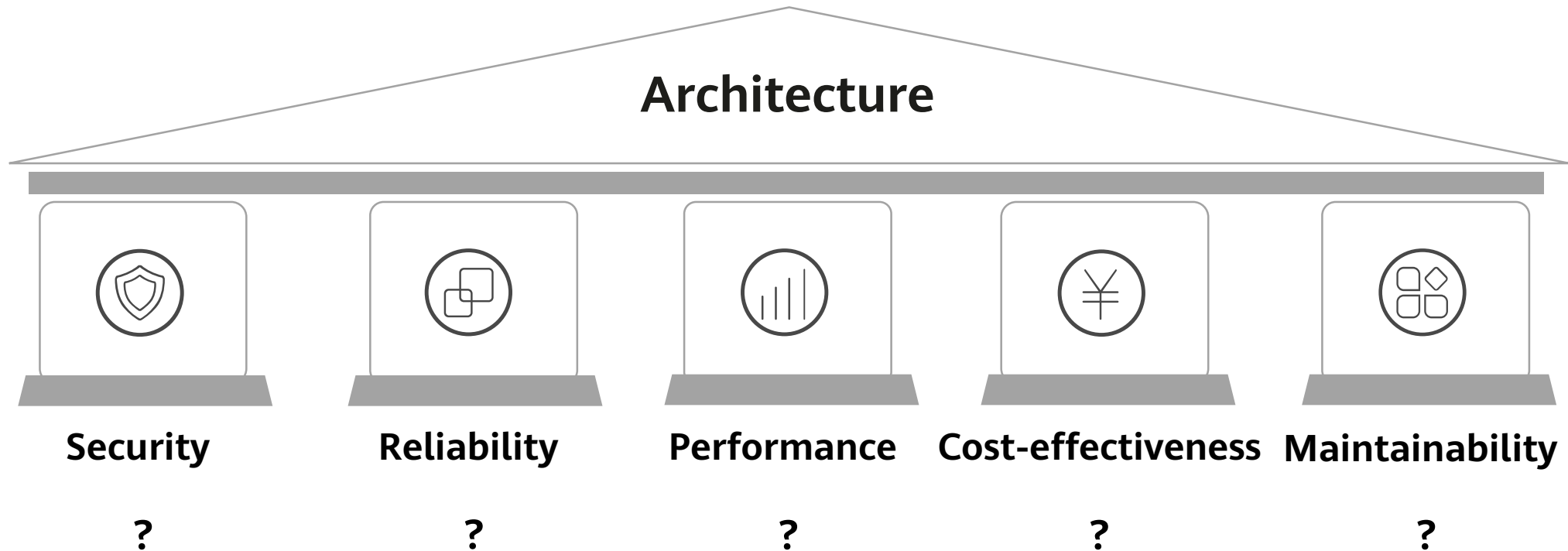
1. Typical Service System Construction on the Cloud
- 2. Discussion on Cloud-based HA Architecture Design**

Discussion

- The preceding architecture implements basic website functions, but still has many problems. What are these problems?

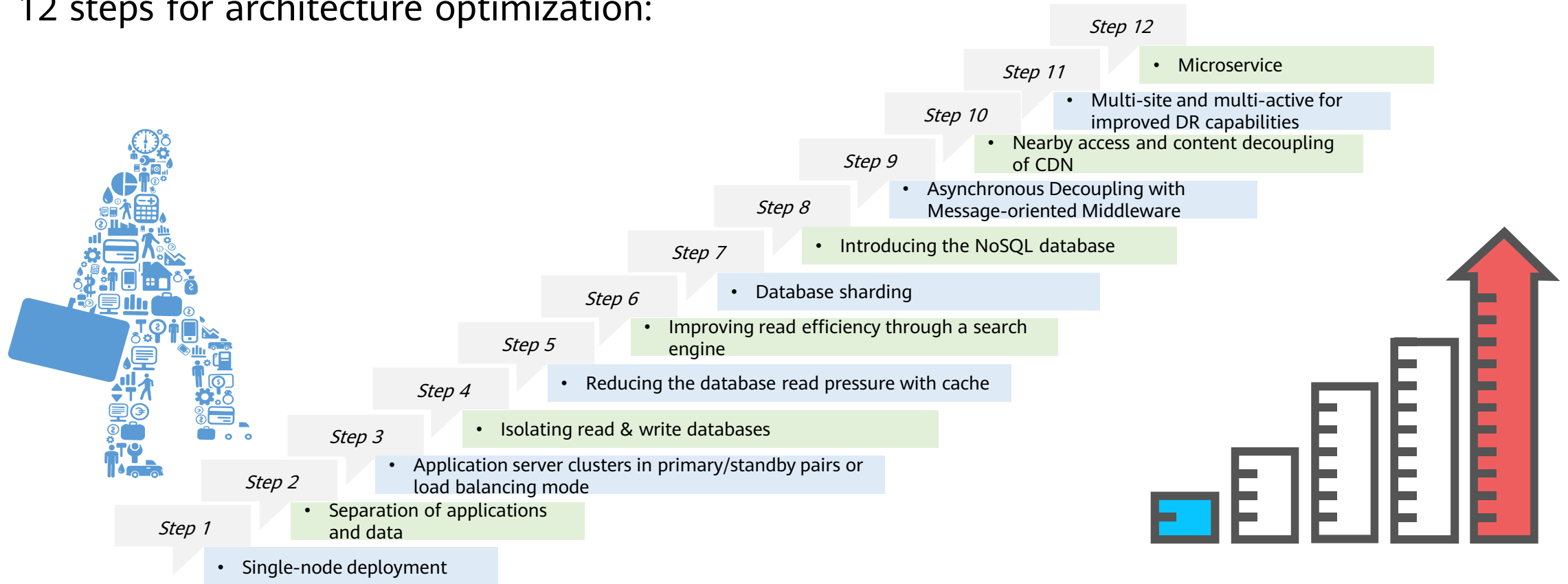


Discussion: From the Perspective of the Five Principles



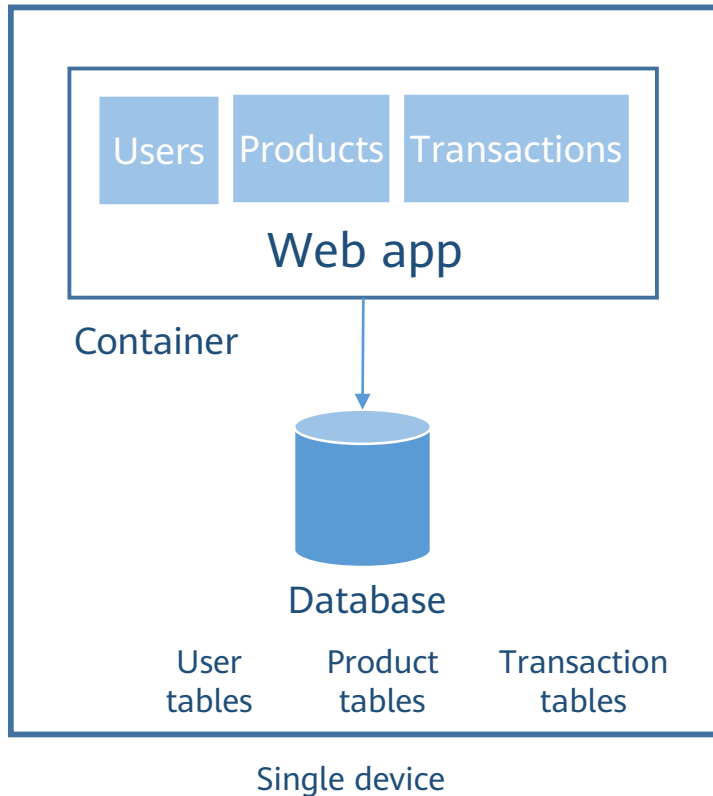
Optimization and Evolution Roadmap for Service Architectures Deployed on Huawei Cloud

- 12 steps for architecture optimization:



Note: The 12 steps are not necessarily implemented in this order, and not every step is mandatory. You need to flexibly perform the steps as needed.

Step 1: Single-Node Deployment



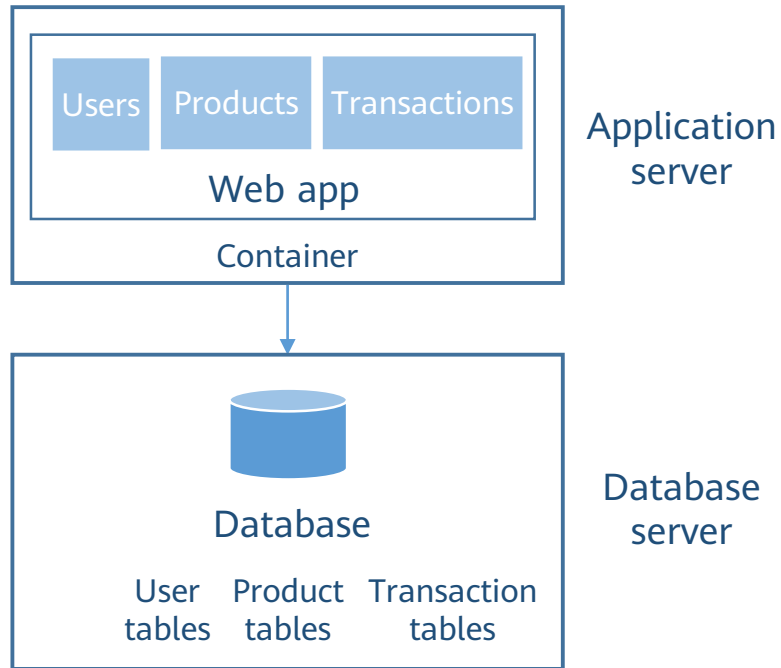
At the early stages of website development, we often run all our software on a single-node system. We deploy resources in JVM containers, such as Tomcat, Jetty, and JBoss, and use JSP/Servlet or open-source frameworks like Maven + Spring + Struts + Hibernate or Maven + Spring + Spring MVC + Mybatis to run our applications. Then, we choose a database management system, such as MySQL, SQL Server, or Oracle, to store data. The database can be accessed and operated through JDBC.

All the preceding software, including the database and application, is installed on one computer. A system is formed.

Huawei Cloud service: ECS

Disadvantages: Resource contention issues occur between applications and databases. A single-node system cannot meet service requirements.

Step 2: Separation of Applications and Data



The **database server** is kept separate from the **web server**. In this way, both the load capacity and disaster recovery capability are improved.

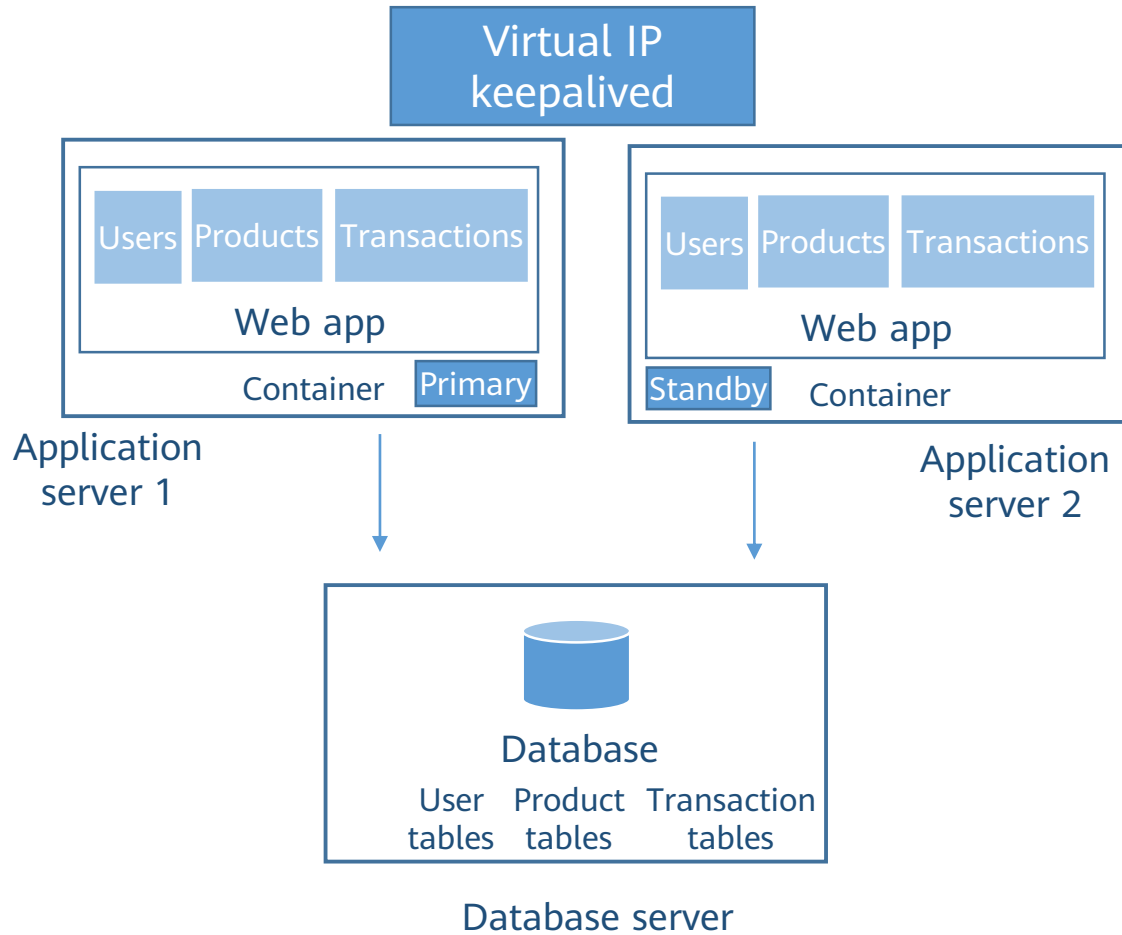
Huawei Cloud services:

ECS (applications) + ECS (self-managed databases)

or **ECS + RDS**

Disadvantages: A single application server cannot handle the increasing access traffic. Single points of failure (SPOFs) may occur.

Step 3: Application Server Clusters in Primary/Standby Pairs



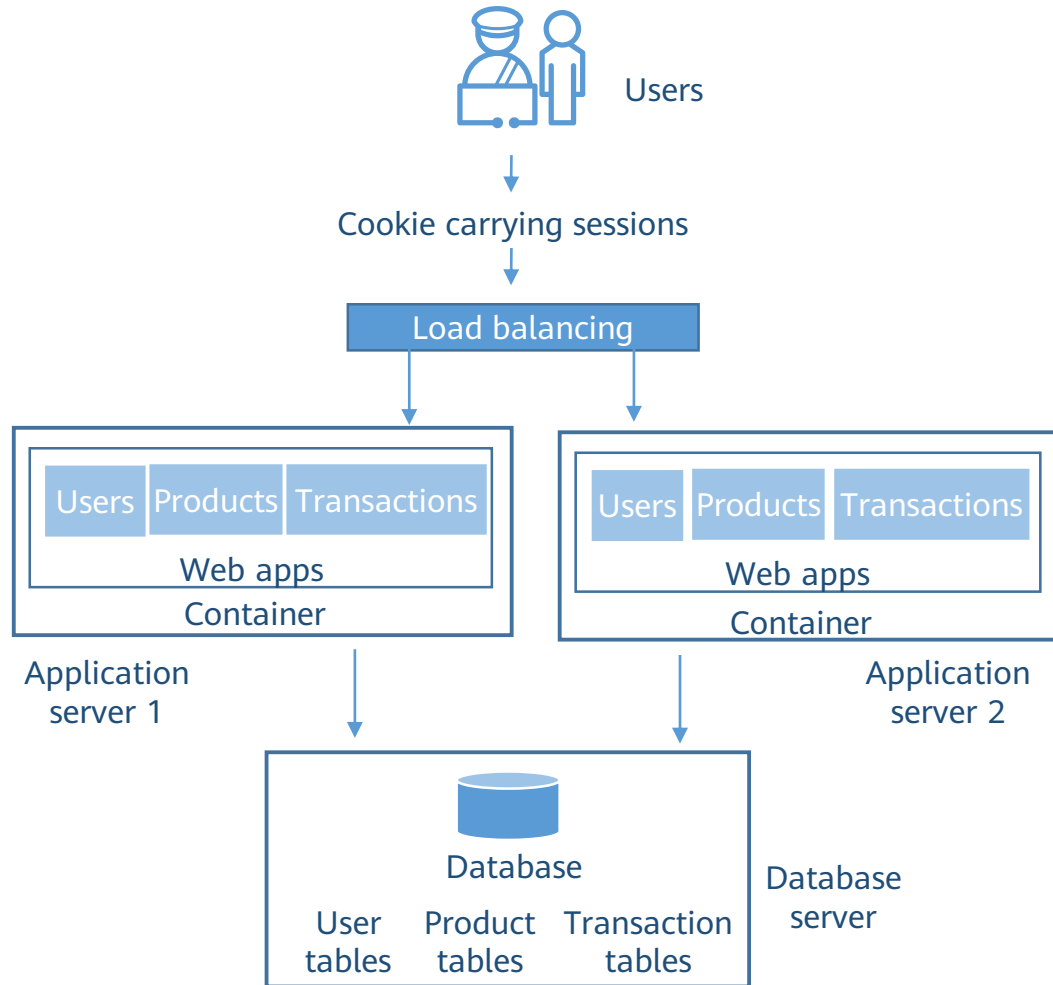
Assume that the database server has sufficient capacity to process all the requests it receives. To eliminate SPOFs, you can deploy **two or more** application servers working in primary/standby pairs to provide services for external systems. These application servers can share the same database. KeepAlive is used for the switchover between primary and standby applications servers.

Huawei Cloud services: ECS + RDS

Features: The primary and standby VMs are associated with the same virtual IP.

Disadvantages: Load balancing cannot be implemented among multiple servers. The servers can work only in primary/standby pairs. The performance of a single server is limited.

Step 3: Application Server Clusters in Load Balancing Mode



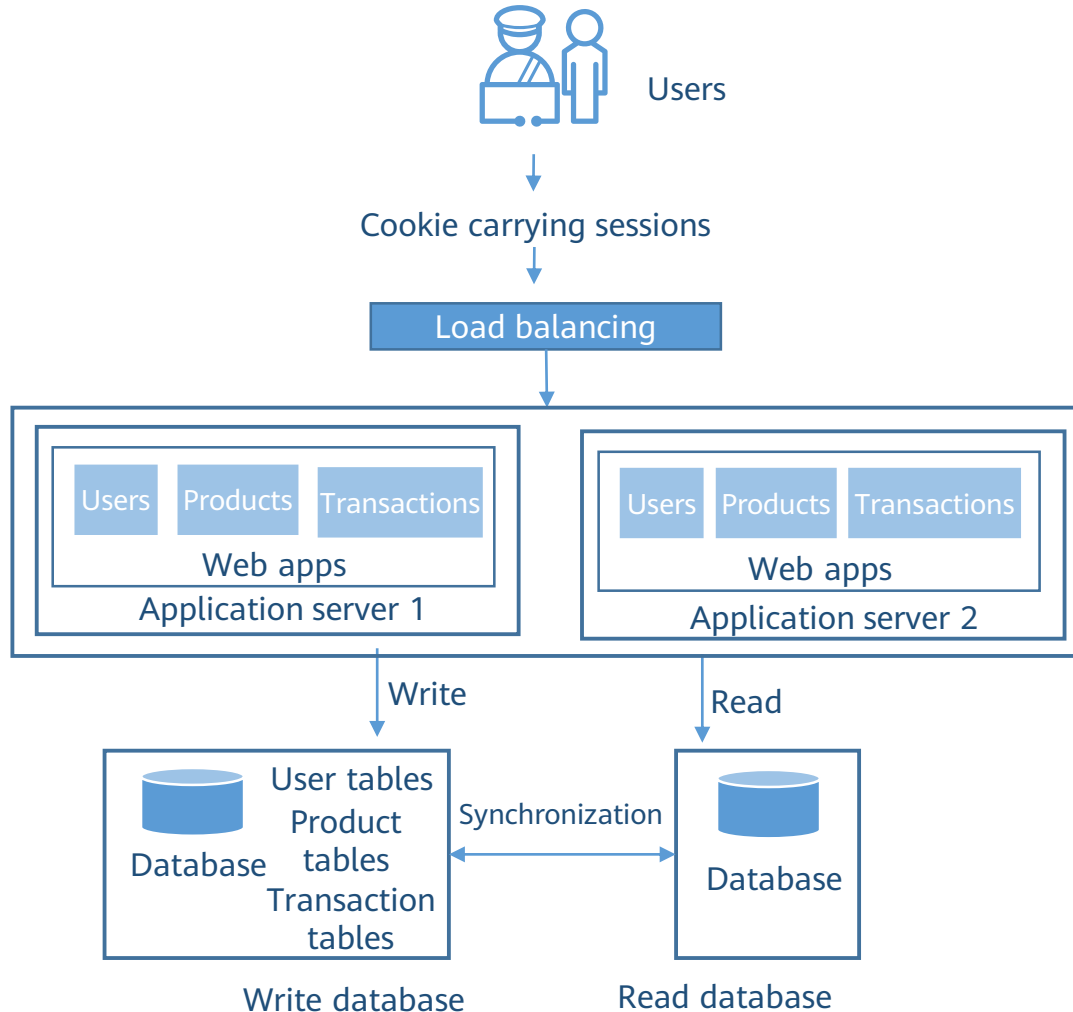
Assume that the database server has sufficient capacity to process data requests. To eliminate SPOFs, you can deploy **two or more** application servers and use Elastic Load Balance (ELB) to distribute traffic across these servers based on the load balancing algorithm you configure. These application servers can share the same database.

Huawei Cloud services: ECS + RDS + ELB (Layer 7 load balancing)

Advantages: ELB greatly improves the availability and scalability of your applications.

Disadvantages: Additional session data needs to be processed. Database loads will still be high.

Step 4: Isolating Read & Write Databases



We always assume that the database load is normal. However, as the access traffic increases, the database load increases gradually. In this case, can we just split the database into two parts for load balancing?

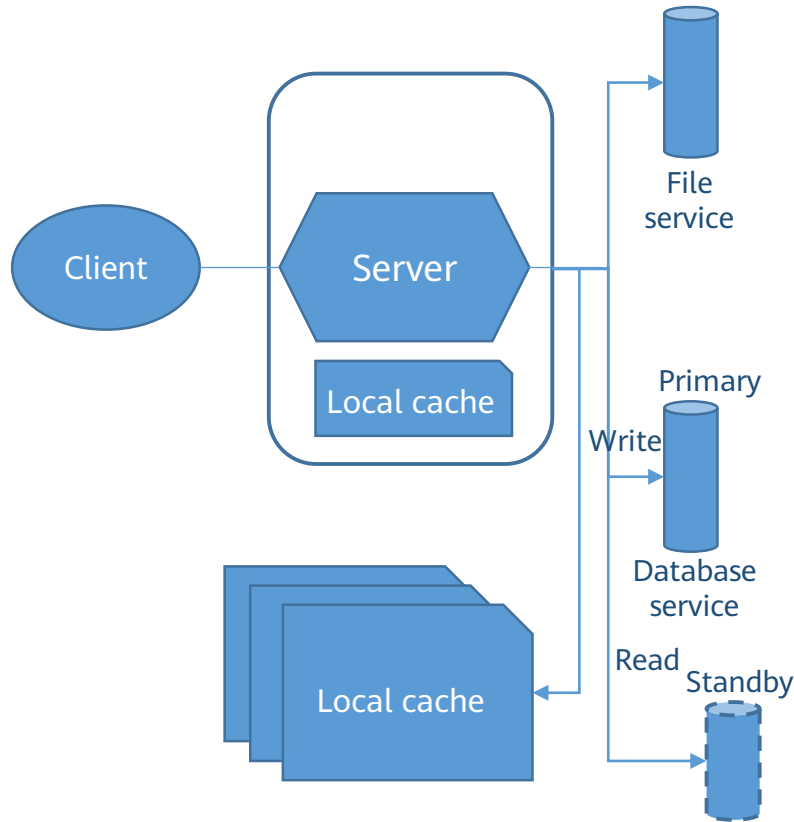
For a database, it's not that simple. If we simply split the database into two parts and distribute requests to database A and database B, the data in the two databases will be inconsistent. In this case, **read/write isolation** and **primary/standby replication** can be used first.

Huawei Cloud services: ECS + RDS (primary/standby) + ELB

Advantages: Read operations are isolated from write operations through primary/standby database replication. Read operations are all moved to the standby database, which reduces pressure on the primary database.

Disadvantages: The application itself needs to select data sources for multiple standby databases.

Step 5: Reducing the Database Read Pressure with Cache



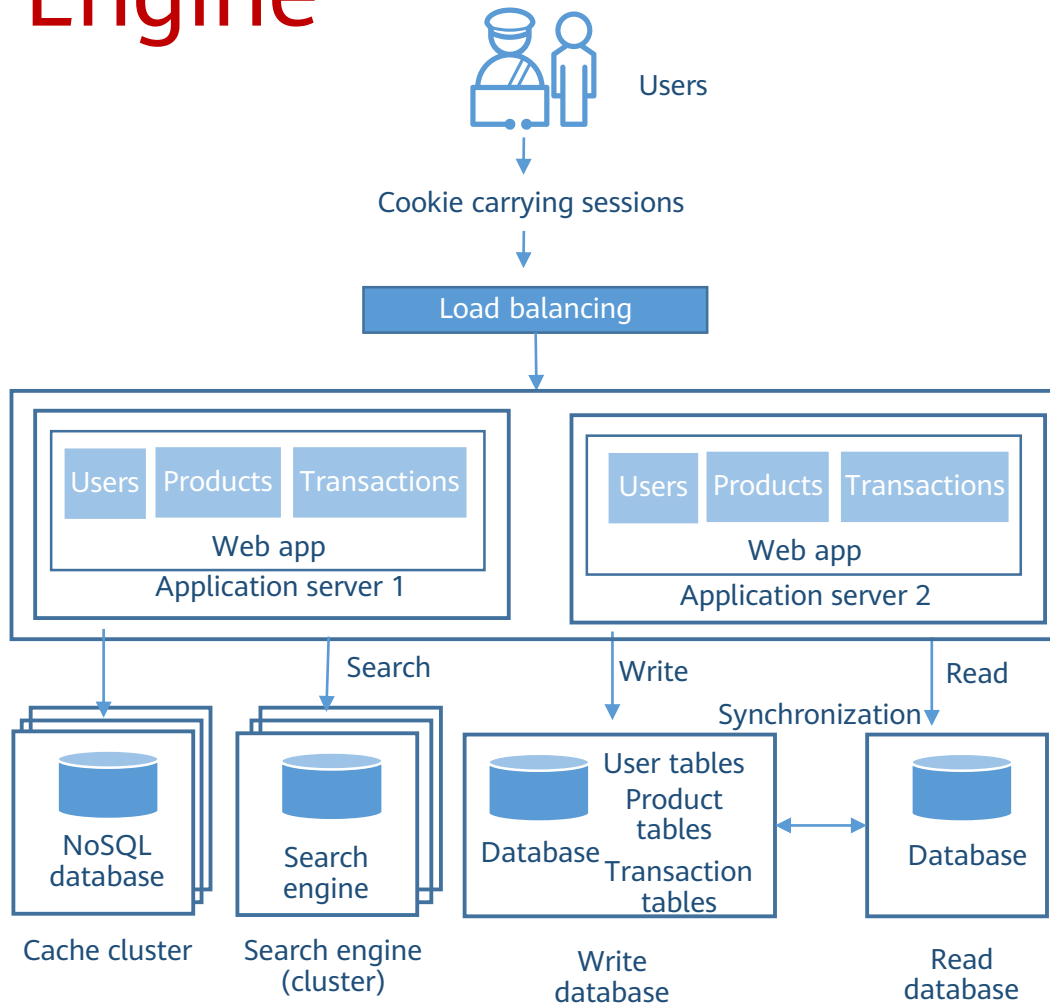
The 80/20 rule is for data access scenarios, where a database only has a small amount of rarely updated yet frequently accessed data. In this case, a cache layer would reduce database access pressure and accelerate data access. The complexity of caching has increased – from monolithic to cluster, and from single point to high availability. The local cache provides quick access but not sharing or high availability. By contrast, a cache cluster provides high-availability, shareable, and large-capacity storage to reduce read and write I/O and speed up system responses.

Huawei Cloud services: ECS + RDS (primary/standby) + ELB + Distributed Cache Service (DCS)

Advantage: Caching hotspot data in DCS removes the need to read the database each time. Reduced read operations lead to better user experience.

Disadvantage: Inefficient fuzzy keyword search

Step 6: Improving Read Efficiency Through a Search Engine



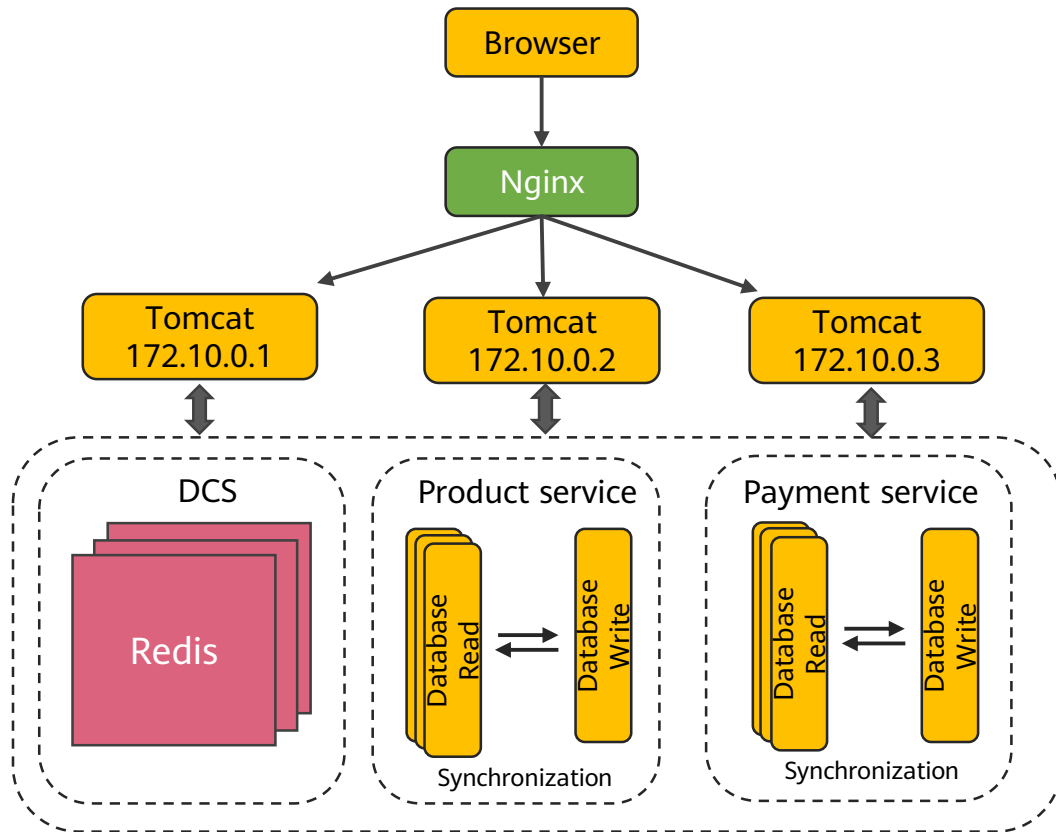
Generally, the read database requires robust **fuzzy search capabilities** even if it is isolated from the write database. In the example of a transaction website, information of released products are stored in databases. The most commonly used operation is to search for products by title. SQL LIKE is used to fulfill this requirement, which is costly and the result is inaccurate. In this case, we can use the **inverted index** of the **search engine** to complete the search.

Huawei Cloud services: ECS + RDS (primary/standby) + ELB + DCS + Elasticsearch (ES)

Advantages: The search speed and accuracy can be **greatly improved**, which makes read operations accurate and efficient and **improves user experience**.

Disadvantages: The write operation pressure on the database is high.

Step 7: Database Sharding - Vertical Partitioning



As data volumes rapidly grow, performance **bottlenecks** become more obvious. To address this issue, we provide two methods: **vertical partitioning** and **horizontal partitioning**.

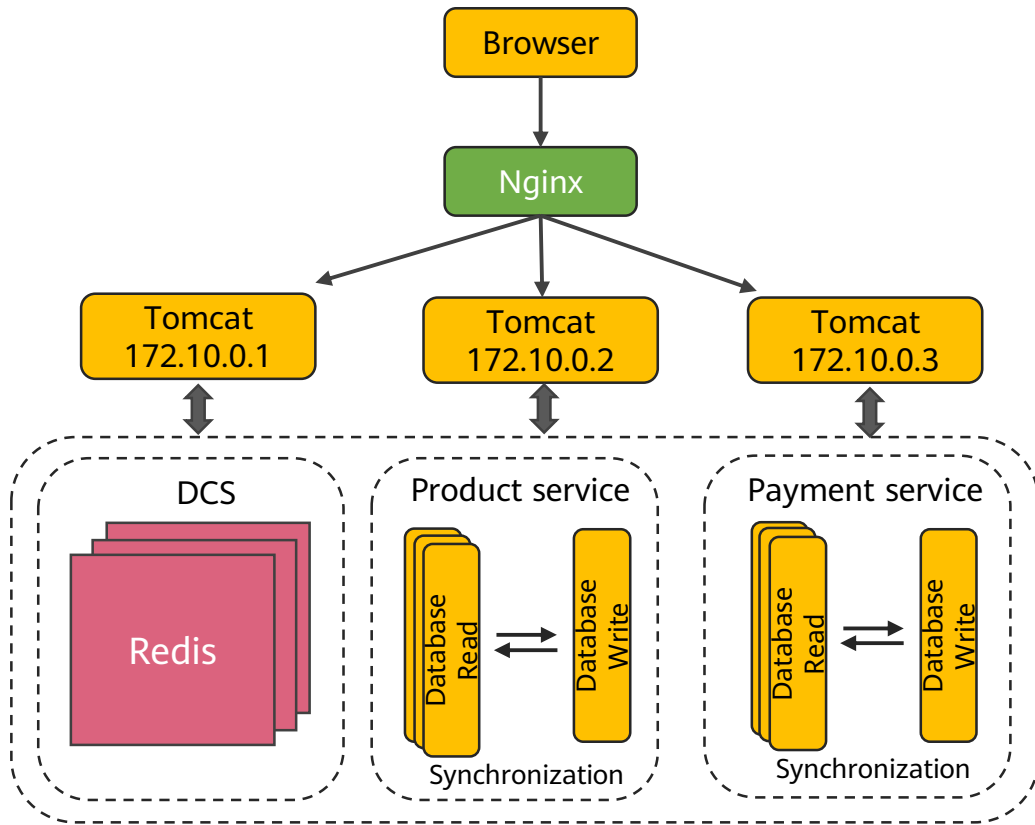
Vertical partitioning allows service data in a database to be split by its function. In this example, transaction data, product data, and user data will be split into different databases.

Huawei Cloud services: ECS + RDS (primary/standby) + ELB + DCS + Elasticsearch (ES) + Distributed Database Middleware (DDM)

Advantages: There is no need to store all service data in the same database. Services can be optimized as needed.

Issues to be addressed: The status consistency and data synchronization among multiple databases need to be maintained. Cross-service transactions and cross-database joins need to be considered.

Step 7: Database Sharding - Horizontal Partitioning



As data volumes rapidly grow, performance **bottlenecks** become more obvious. To address this issue, we provide two methods: **vertical partitioning** and **horizontal partitioning**.

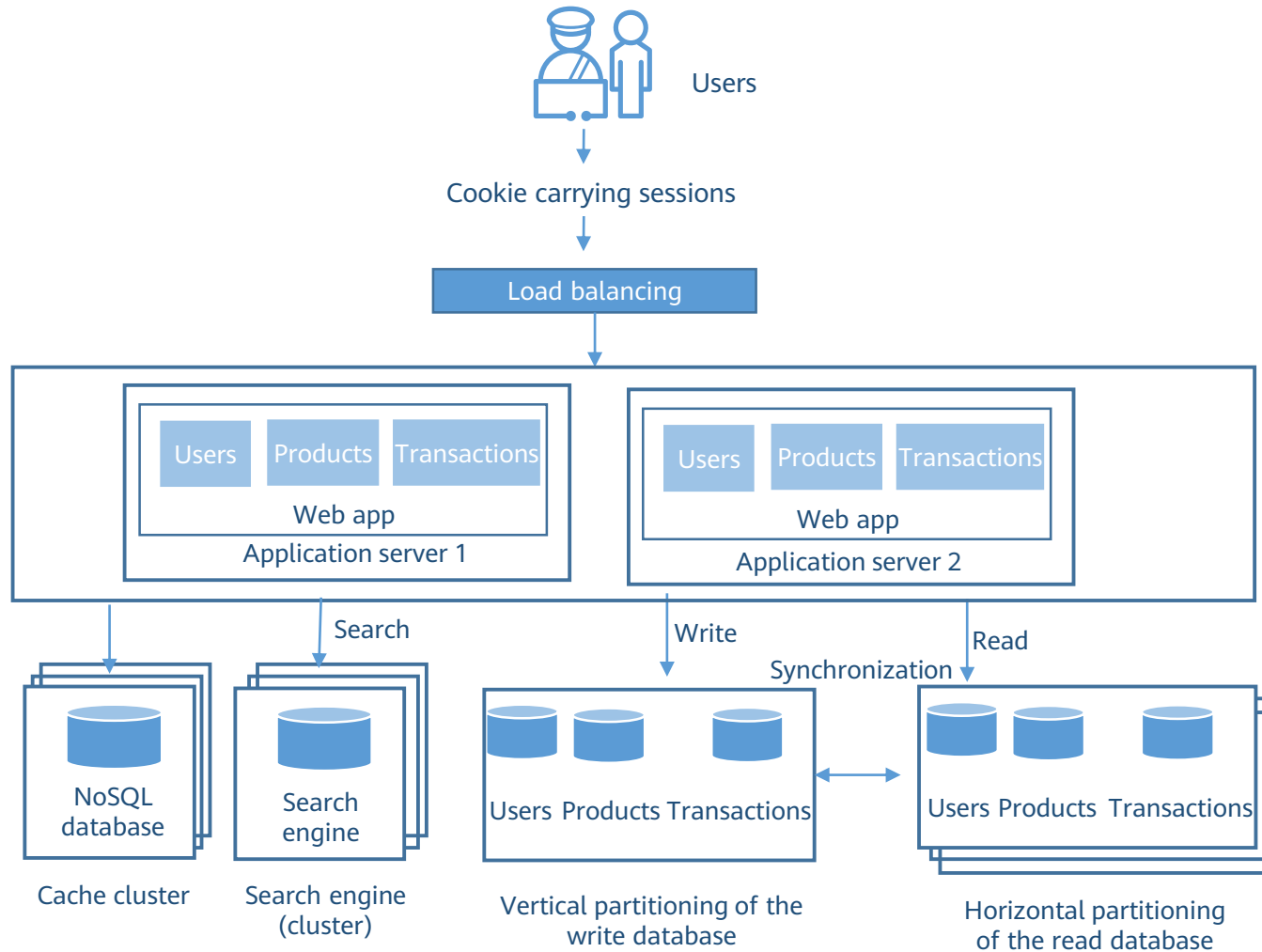
Horizontal partitioning is performed when **the data volume** or updated volume of a service reaches the upper limit of **a single database (table)**. Horizontal partitioning splits the given table into **two or more** databases without having to create separated tables for each database.

Huawei Cloud services: ECS + RDS (primary/standby) + ELB + DCS + Elasticsearch (ES) + DDM

Advantages: This method is a great fit for scenarios where the amount of data to be written increases rapidly.

Issues to be addressed: SQL routing is an issue for applications that access user information. As the user information is stored in two databases, it is necessary to know where the data to be accessed is located. The processing of **primary keys** becomes different. For example, the original **auto-increment field** cannot be used directly now. It is more troublesome when **pagination** query is required.

Step 7: Database Sharding - Overall Architecture

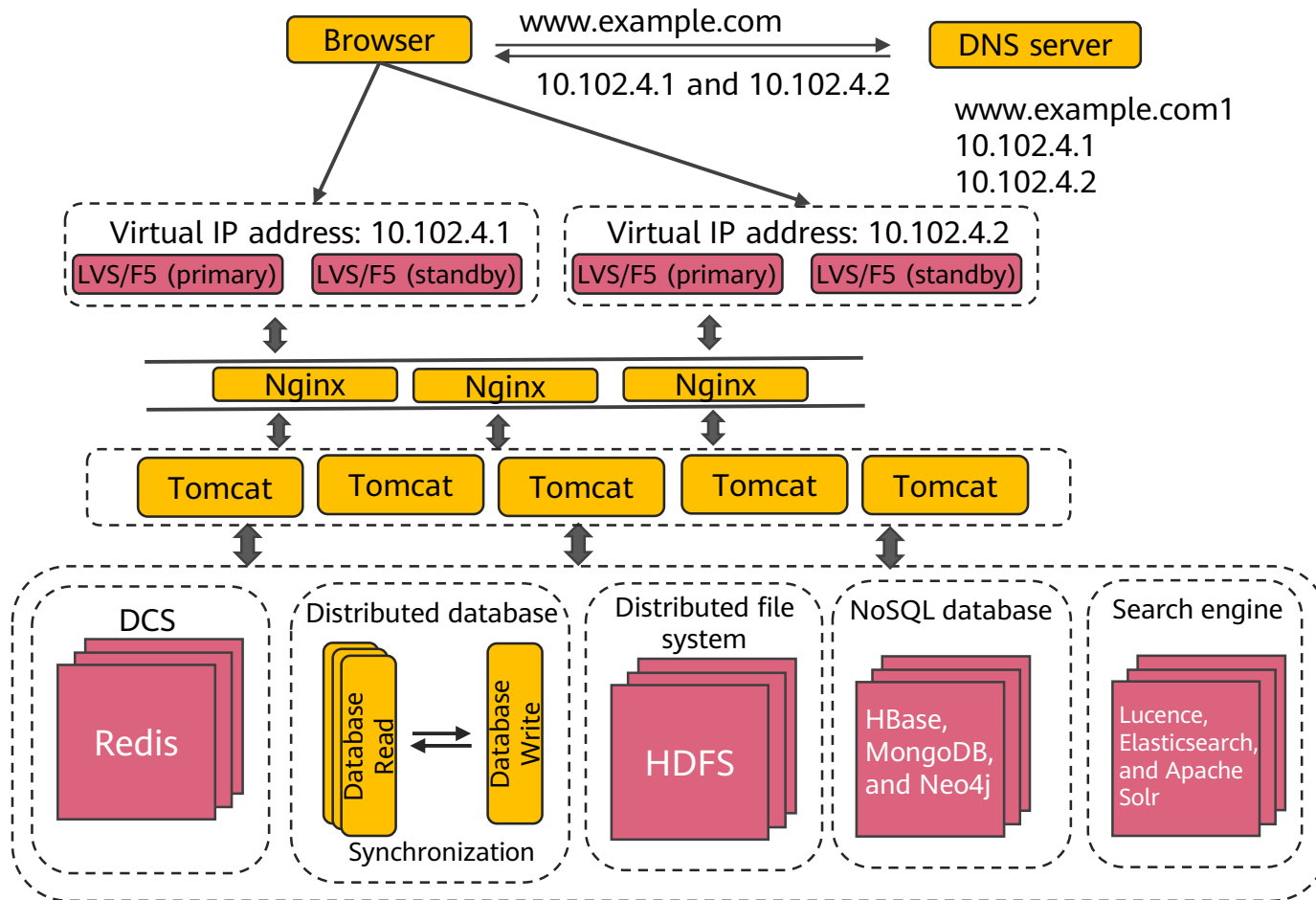


Huawei Cloud services: ECS + RDS (primary/standby) + ELB + DCS + Elasticsearch (ES) + DDM

Disadvantages: All users need to access the central site. For users who are far away from the central site or transmit data across networks, high latency results in poor user experience.

Step 8: Introducing the NoSQL Database

Collecting report statistics may fail when there too much data. In addition, running complex queries can cause other queries to slow down. Databases are not suitable for full-text retrieval and variable data structures. Therefore, different solutions need to be introduced for specific scenarios. For example, MRS HDFS can be used to store massive numbers of files, MRS HBase and Redis can be used to store key-value data, and OLAP can be used for multidimensional analysis.

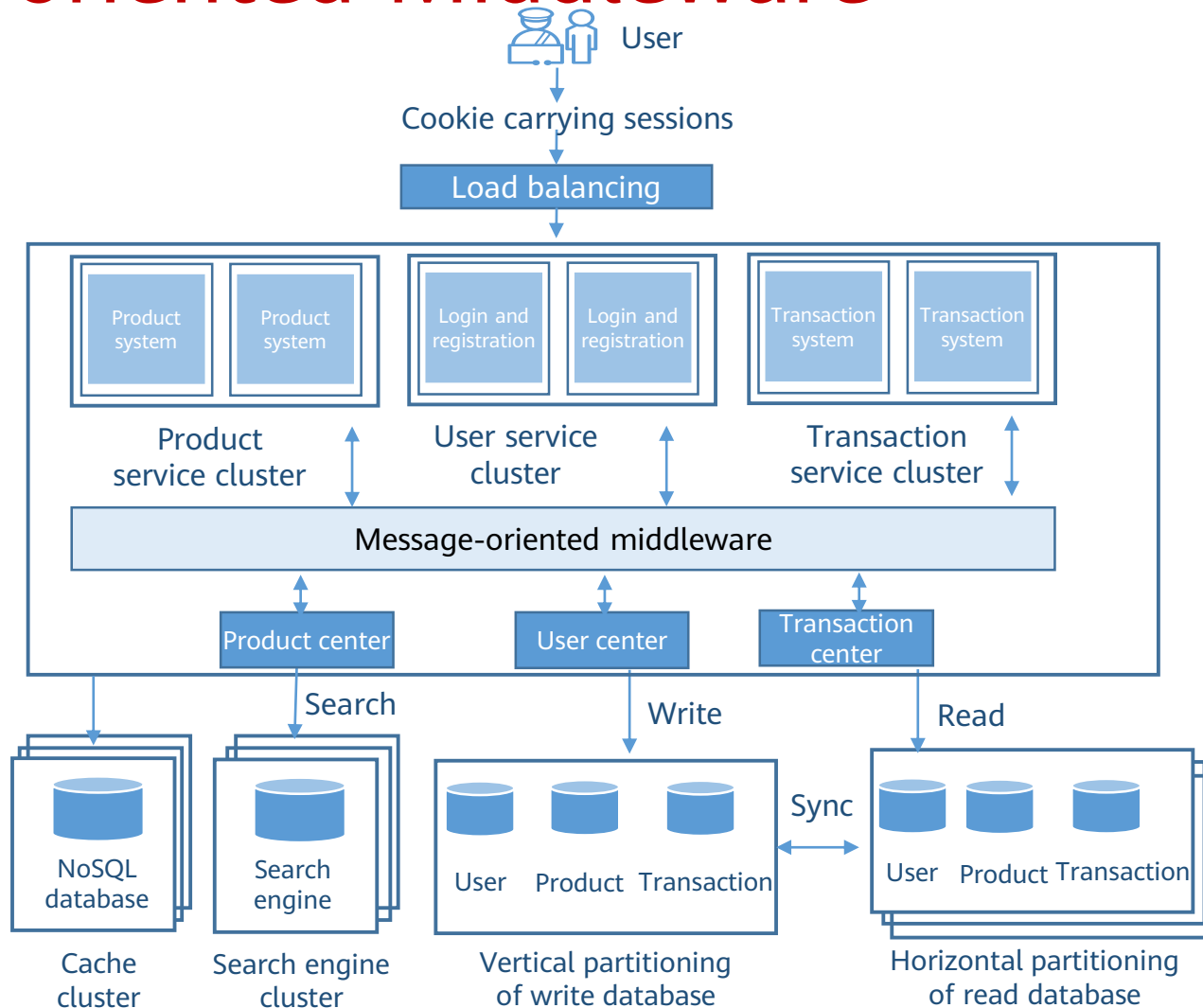


Huawei Cloud services: MapReduce Service (big data) + Data Warehouse Service (DWS) + Document Database Service (DDS)

Advantages: This architecture is compatible with the industry ecosystem and provides enhanced security, usability, and high availability.

Issues to be addressed: More components will be introduced to meet various requirements and expand service dimensions. However, service upgrade and iteration become more difficult because every application contains too much service code.

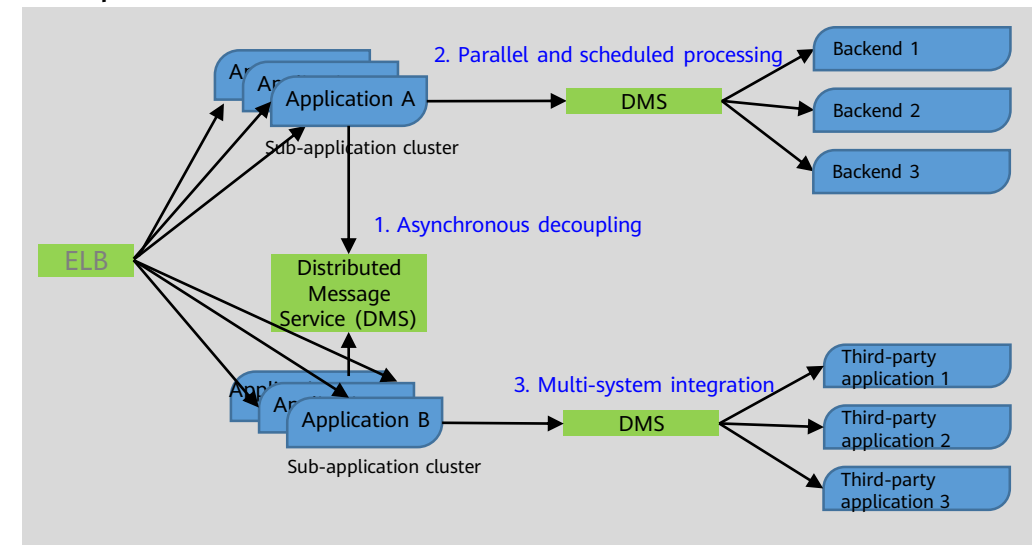
Step 9: Asynchronous Decoupling with Message-oriented Middleware



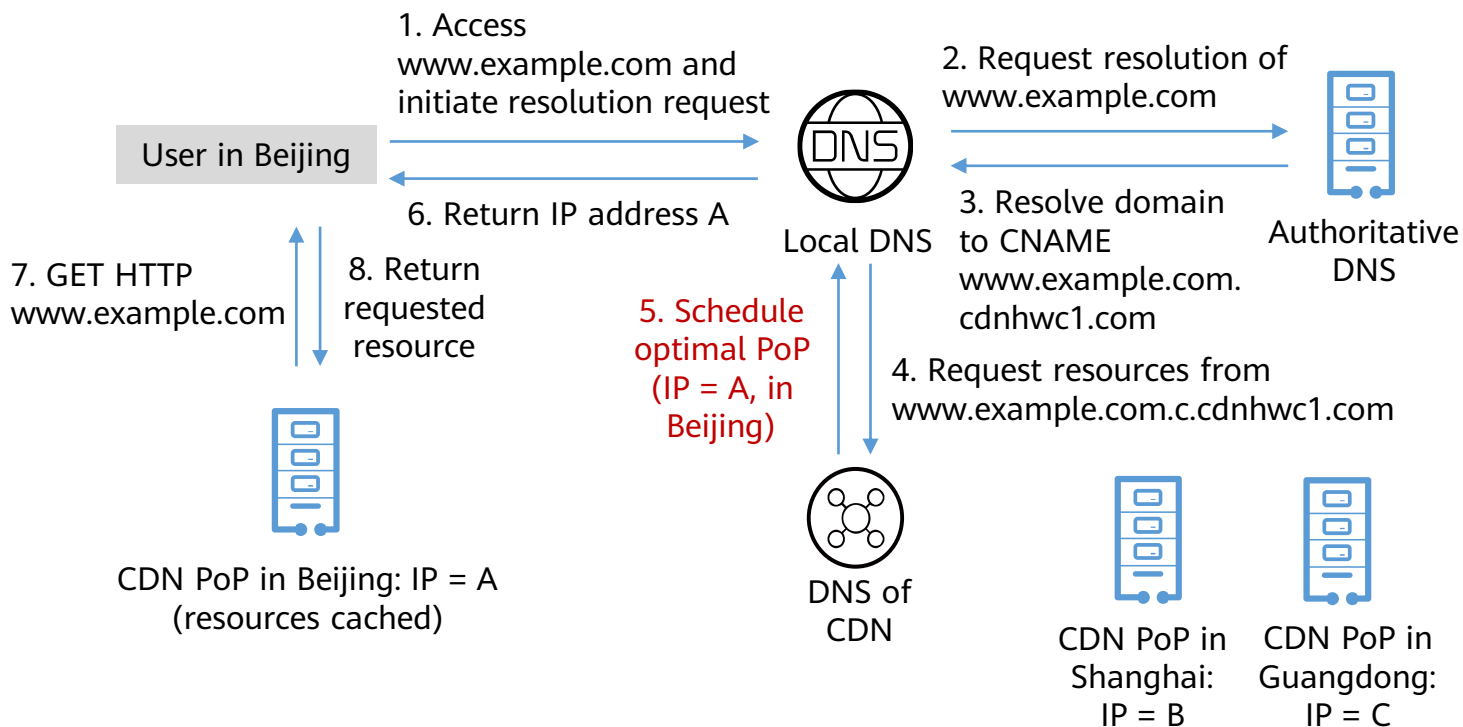
A fully managed, powerful message queuing service for reliable, flexible, and asynchronous communication between distributed applications. It supports HTTP APIs, TCP SDK, and Kafka SDK for data access.

Issues to be addressed:

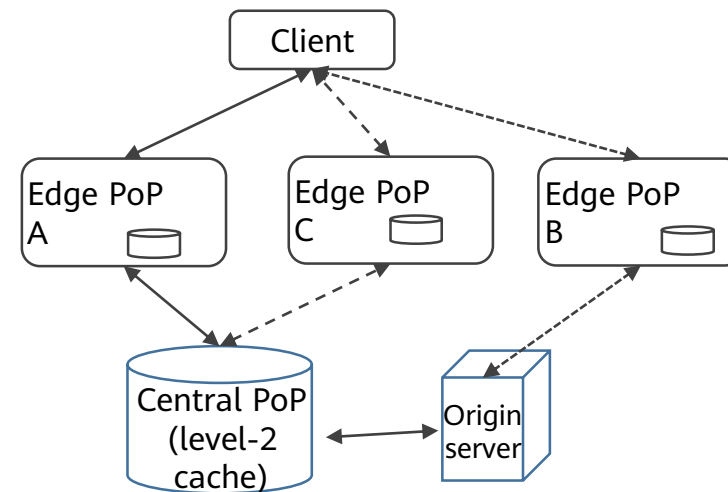
Sub-systems are complex and tightly coupled, making development and maintenance difficult.



Step 10: Nearby Access and Content Decoupling



As businesses grow, the pressure on servers increases. Dynamic/static content decoupling helps reduce the pressure. This technology separates static resources of services from backend applications to speed up static resource access. Static resources are cached on CDN and deployed in equipment rooms of network providers. With CDN, users obtain the static resources from the nearest equipment room.

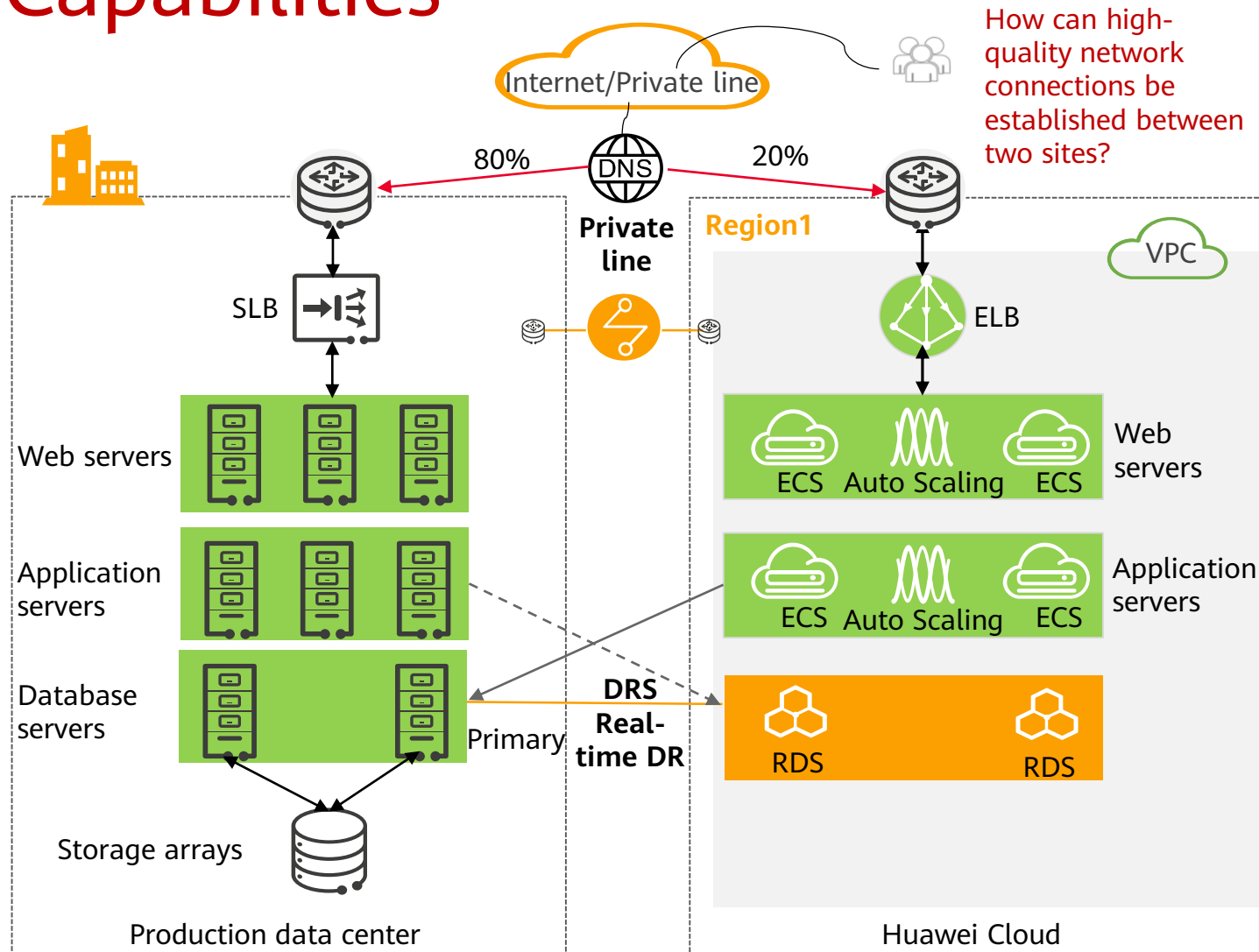


Huawei Cloud service: Content Delivery Network (CDN)

Advantages: Backend applications are more service-oriented and only need to provide APIs. The frontend can use better technologies to improve access efficiency.

Issues to be addressed: cache update of static files and update cost of the frontend and backend

Step 11: Multi-site and Multi-active for Improved DR Capabilities



As the number of users increases, a system deployed on a single site fails to handle large-scale accesses. When regional DR cannot be deployed, a DR site is required to improve the system concurrency capability, and the system availability can be improved using active-active DR.

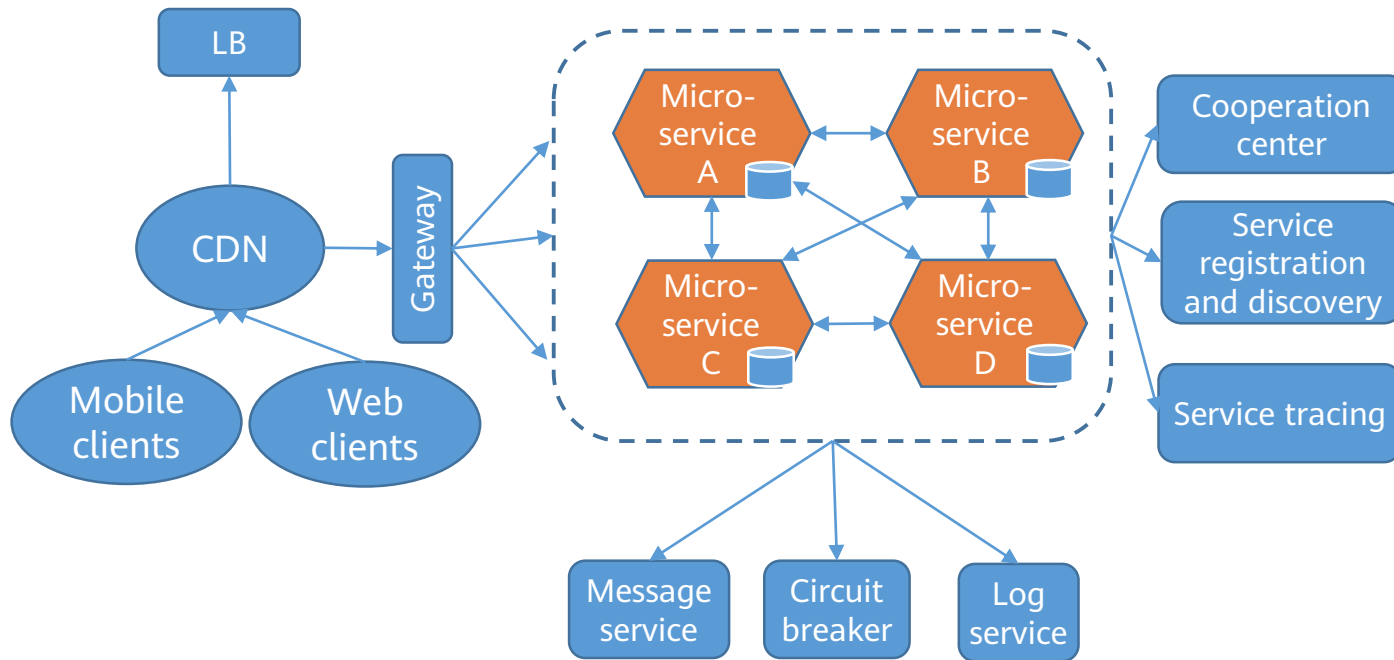
Huawei Cloud services: ELB (both layer 4 and layer 7 support) + CBR + RDS (primary/standby)

Third-party: Intelligent DNS, GSLB, and application-level DR solution

Advantages: Single-site access pressure is reduced and system availability improved.

Disadvantages: As the data volume grows and services develop, there will be more retrieval and analytics requirements, relational databases alone may fail to meet all the requirements.

Step 12: Microservice - Architecture



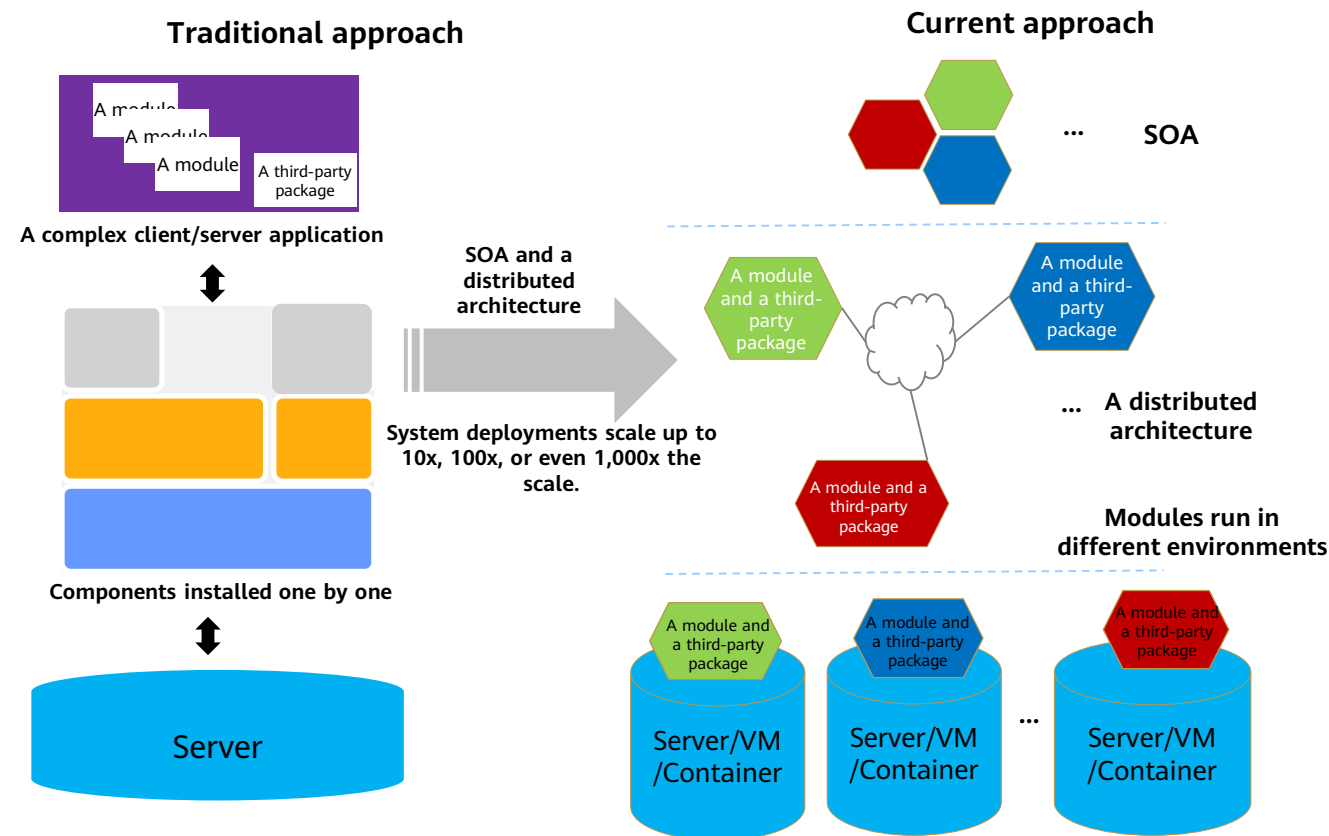
In a microservice architecture, services split into components that quickly assemble unto a system. Similar components are also independently deployed for flexibility.

However, the more distributed services there are, the more problems: registration and discovery, unified access and permission control, load balancing, decentralized configuration management, circuit breaker, and monitoring.

FusionStage introduces other components in response: registration and discovery, gateway, load balancing, tracing, centralized configuration, and circuit breaker components.

Huawei Cloud service: FusionStage (distributed and microservice development framework)

Step 12: Microservice - Containerization

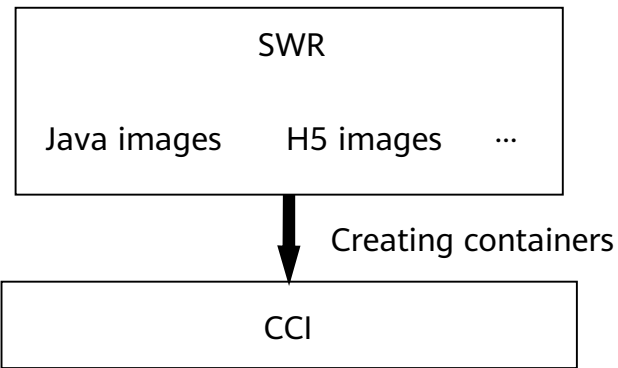


Supported by Huawei Cloud Container Engine (CCE) and Cloud Container Instance (CCI)

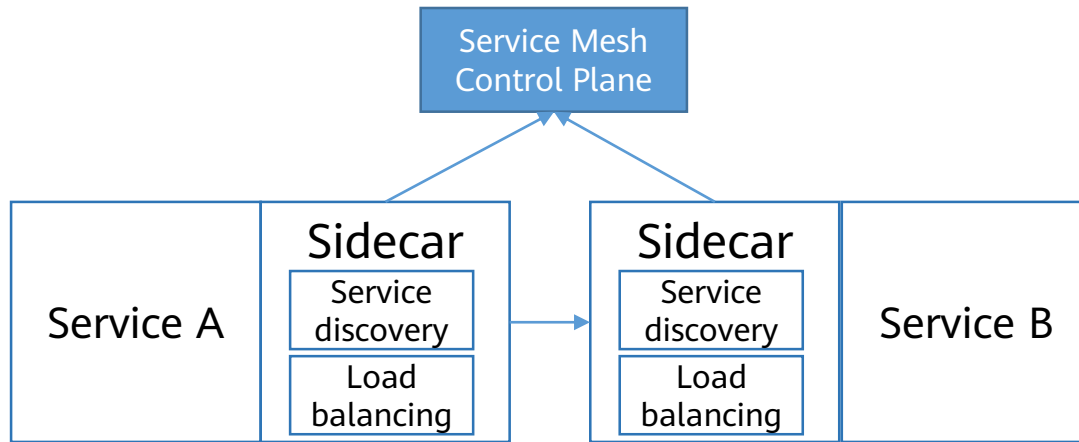
Docker is a tool designed to solve problems related to large-scale system deployment. Docker is much more efficient than traditional deployment.

In addition to a mature framework, a tool for automatically deploying microservices is needed. Containers have become an ideal approach. Docker and other container technologies make it easy to develop and continuously deliver software. They have greatly simplified microservice O&M and significantly reduced the risk of system breakdowns.

Docker first packages the entire OS into an artifact in the form of image and then ships this image to a host that your services are deployed on. You can directly start the Docker image to run your services. With Docker, service deployment and O&M are easy.



Step 12: Microservice - Service Meshes



In Huawei Cloud, ASM (that provides Istio-based service meshes) addresses major challenges faced by ServiceCombo (a SDK-based microservice governance framework).

While a microservice architecture enables rapid iteration and continuous delivery, it also adds complexity. In addition to the business logic, developers have to invest time into network functions such as service registration, service discovery, service communication, load balancing, circuit breaking, and timeouts. Most third-party libraries or components, such as Hystrix and Eureka, provide these network functions. However, these libraries or components are intrusive. Developers need to consider the non-business logic when using these libraries or components. That's where service meshes come into the picture.

When it comes to the realization of a microservice architecture, a service mesh serves as an infrastructure layer that handles all communications between proxies for message transfer on demand. This gives developers more freedom to focus more on the business logic and less on network functions. In addition, service meshes are transparent to services and are always deployed with services.

Quiz

(Multiple-choice question) Which of the following statements about Huawei Cloud infrastructure service features are false?

- A. A VPC is a logically isolated network. By default, VPCs can interconnect with each other, and subnets in a VPC are isolated from each other.
- B. SFS provides file storage services on the cloud and has good cross-region read and write performance.
- C. RDS is a one-stop data service provided by Huawei Cloud and has good cross-AZ HA capabilities.
- D. IAM is a permissions management service provided by Huawei Cloud to authorize cross-account access.



Summary

- This lesson described Huawei Cloud infrastructure services and how to improve the performance of cloud service systems from the perspective of the five principles.



Acronyms and Abbreviations

- ECS: Elastic Cloud Server
- EVS: Elastic Volume Service
- VPC: Virtual Private Cloud
- EIP: Elastic IP
- RDS: Relational Database Service
- OBS: Object Storage Service
- CTS: Cloud Trace Service

Thank You.

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